Lecture 2 v2:
Review of Statistics

25117 - Econometrics

Universitat Pompeu Fabra

September 30th, 2024

What we learned in the last lesson

- Econometrics quantifies economic relationships
- Randomized control experiments are often too expensive or unethical. We have to rely on observational data that is potentially jointly determined.
- The probabilities with which a random variable takes on different values are summarized by the cumulative and probability distribution functions
- The expected value of a random variable Y, $\mathbb{E}(Y)$, is its probability-weighted average value. The variance is $\mathbb{E}[(Y - \mathbb{E}(Y))^2]$.
- The joint probabilities for two random variables, X and Y, are summarized by their joint probability distribution. The conditional probability distribution of Y given $X = x$ is the probability distribution of Y, conditional on X taking on the value x.

What we learned in the last lesson

- Simple random sampling produces n random observations that are i.i.d.
- The sample average, $\overline{Y}$, is a random variable and an estimator of μ_Y before the sample is observed. If observations are i.i.d. then,
  1. $\overline{Y}$ has mean μ_Y and variance $\sigma^2_{\overline{Y}} = \frac{\sigma^2_Y}{n}$;
  2. the LLN says that $\overline{Y}$ converges in probability to μ_Y
  3. the CLT says that $(\overline{Y} - \mu_Y)/\sigma^2_{\overline{Y}} \sim \mathcal{N}(0;1)$ when n is large
- After observing the sample, the realized value $\overline{y}$ is the estimate produced by the estimator $\overline{Y}$.
- The sample average, $\overline{Y}$ is a BLUE estimator of the population mean. If observations are i.i.d. then,
  1. $\overline{Y}$ is unbiased;
  2. by the LLN, $\overline{Y}$ is consistent
  3. by the CLT, $\overline{Y}$ has an approximate normal sampling distribution when n is large
  4. $\overline{Y}$ is efficient – it has the smallest sampling variance of all linear unbiased estimator

What is Hypothesis Testing?

Hypothesis Testing is a statistical method used to make decisions about population parameters based on sample data.

- We start with two competing hypotheses:
  - Null Hypothesis (H_0): The default assumption, no effect, no difference, etc.
  - Alternative Hypothesis (H_1): The statement we want to test, suggests an effect or difference.
- We collect sample data and use it to make an informed decision about which hypothesis is more likely.
- For example, if we want to test that a population mean takes a specific value, we write $H_0 : \mu_Y = \mu_Y^0$
- Here the alternative hypothesis would be $H_1 : \mu_Y \neq \mu_Y^0$. We talk about two-sided alternatives because μ_Y may be greater or smaller than μ_Y^0

Why Use Hypothesis Testing?

Hypothesis testing helps us:

- Make data-driven decisions.
- Evaluate claims or theories.
- Determine if observed differences are statistically significant or just due to chance.
- Assess the reliability of our results.

It provides a structured approach for drawing conclusions from data.

Example: A pharmaceutical company develops a new drug to lower blood pressure.

- Hypotheses:
  - Null Hypothesis (H_0): The new drug has no effect on blood pressure.
  - Alternative Hypothesis (H_1): The new drug is effective in lowering blood pressure.
- Clinical trial data is collected to assess the drug's efficacy.
- Hypothesis testing determines if the observed blood pressure reduction is statistically significant.
- Outcome: Approval for medical use if the null hypothesis is rejected.

Key Concepts

- **Null Hypothesis (H_0)**: The default hypothesis, often representing no effect or no difference.
- **Alternative Hypothesis (H_1)**: The statement we want to test, suggesting an effect or difference.
- **Significance Level (α)**: The threshold for determining statistical significance. Common choices are 0.05 or 0.01.
- **Test Statistic**: A numerical summary of the data used in hypothesis testing.
- **P-value**: The probability of obtaining results as extreme as the observed data, assuming the null hypothesis is true.

These concepts are fundamental in hypothesis testing and guide our decision-making process.

P-Value for a One-Sample Mean Test

- We test a population mean against a hypothesized value:

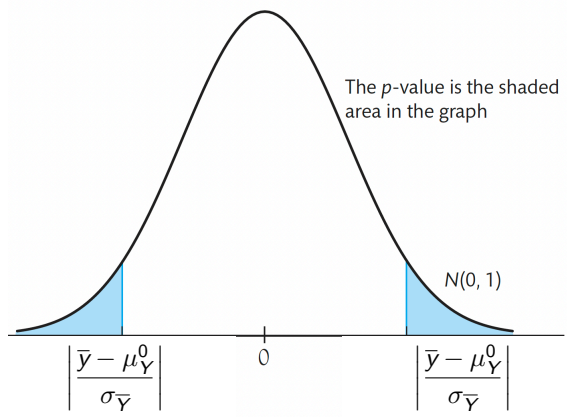
$$H_0 : \mu_Y = \mu_Y^0 \qquad \text{vs.} \qquad H_1 : \mu_Y \neq \mu_Y^0$$

- This is a two-sided one-sample test.
- The p-value is the probability, assuming H_0 is true, of obtaining a sample mean at least as far from μ_Y^0 as the observed value $\bar{y}$:

$$Pr_{H_0}[|\ \overline{Y} - \mu_Y^0\ | > |\ \bar{y} - \mu_Y^0\ |]$$

- Computing the p-value therefore requires the sampling distribution of the random variable $\overline{Y}$.
- For large n, the standardized mean is approximately normal under H_0:

$$\frac{\overline{Y} - \mu_Y^0}{\sigma_{\overline{Y}}} \approx \mathcal{N}(0,1).$$



Sample Variance and Sample Standard Deviation

- The sample variance, s_Y^2 is an estimator of the population variance σ_Y^2

$$s_Y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \overline{Y})^2$$

- The sample standard deviation, $s_Y = \sqrt{s_Y^2}$, is an estimator of the population standard deviation σ_Y
- There are two differences with the population variance, $\mathbb{E}[(Y-\mu_Y)^2]$:
  1. μ_Y is unknown, so it is replaced by its natural estimator $\overline{Y}$
  2. We divide by $n-1$ (instead of n) to correct for the small downward bias generated by estimating μ_Y with $\overline{Y}$ (Bessel's correction)
- We can show that $s_Y^2 \xrightarrow{p} \sigma_Y^2$ as $n \to \infty$

Step 1: The Standard Deviation of the Sample Mean

- For i.i.d. observations, the sample mean $\overline{Y}$ has variance

$$\sigma^2_{\overline{Y}} = \frac{\sigma^2_Y}{n}.$$

- Taking the square root gives its theoretical standard deviation:

$$\sigma_{\overline{Y}} = \frac{\sigma_Y}{\sqrt{n}}.$$

- This is smaller when n is larger: averaging more observations makes the sample mean more precise.

Step 2: Estimating the Standard Error

- In practice, the population standard deviation σ_Y is usually unknown.
- We estimate it with the sample standard deviation s_Y:

$$s_Y^2 \xrightarrow{p} \sigma_Y^2.$$

- Replacing σ_Y by s_Y gives the estimated standard deviation of $\overline{Y}$:

$$SE(\overline{Y}) = \widehat{\sigma}_{\overline{Y}} = \frac{s_Y}{\sqrt{n}}.$$

- The standard error measures the sampling uncertainty of the estimator $\overline{Y}$.

Step 3: Using the Standard Error in the P-Value

- **When σ_Y is unknown, use the estimated standard error in the two-sided mean test. Remember: the p-value is a probability.**

$$\text{p-value} = 2\Phi\left(-\left|\frac{\bar{y} - \mu_Y^0}{SE(\bar{Y})}\right|\right)$$

- The numerator is the observed distance from the null value, $\bar{y} - \mu_Y^0$.
- Dividing by the standard error converts this distance into standard-error units.
- The factor 2 accounts for both tails because the alternative is two-sided.
- This uses the assumption that $Y_1, \ldots, Y_n$ are i.i.d. and that the standardized mean is approximately normal.

The function Φ

$\Phi(z)$ is the probability that a standard normal random variable is less than or equal to z.

The Standard Normal Cumulative Distribution Function

- The standard normal cumulative distribution function is

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx.$$

- It gives the area under the standard normal curve to the left of z.
- The p-value uses this area to add the probabilities in both tails.

The t-statistic and its use for hypothesis testing

- The standardized sample average is called t-statistics, t-ratio, or Student's t-statistic.

$$\mathrm{t} = \frac{\overline{y} - \mu_Y^0}{SE(\overline{Y})}$$

- Because $s_Y^2 \xrightarrow{p} \sigma_Y^2$ as $n \to \infty$, then $t \sim \mathcal{N}(0,1)$ as $n \to \infty$
- We choose a significance level α before looking at the result. We reject H_0 (or say that the observed estimate $\overline{y}$ is significantly different from μ_Y^0) when

$$\textbf{p-value} \leq \alpha.$$

  For example, at the 5% level, $\alpha = 0.05$ and we reject when the p-value is at most 0.05.
- To summarize, testing H^0 against H^1 involves:
  1. Compute the standard error of $\overline{Y}$, $SE(\overline{Y})$
  2. Compute the corresponding t-statistic
  3. Compute the p-value.
  4. Under H^0, $t^{obs} \approx \mathcal{N}(0,1)$, so for a two-sided test at the 5% significance level, p-value ≤ 0.05 is equivalent to $\mid t^{obs} \mid \geq 1.96$.
  5. At the 10% significance level, p-value ≤ 0.10 is equivalent to $\mid t^{obs} \mid \geq 1.645$.
- But what are the consequences of making errors in hypothesis testing?

Type I and Type II Errors in Hypothesis Testing

Type I Error (False Positive)

- Occurs when we reject a null hypothesis that is actually true.
- Denoted as α (alpha), the chosen significance level.
- Probability of Type I Error:

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}).$$

- Example: Incorrectly concluding a drug is effective when it's not.
- Consequence: Often considered more serious due to potential harm.

Type II Error (False Negative)

- Occurs when we fail to reject a null hypothesis that is actually false.
- Denoted as β (beta).
- Probability of Type II Error:

$$\beta = P(\text{do not reject } H_0 \mid H_1 \text{ true}).$$

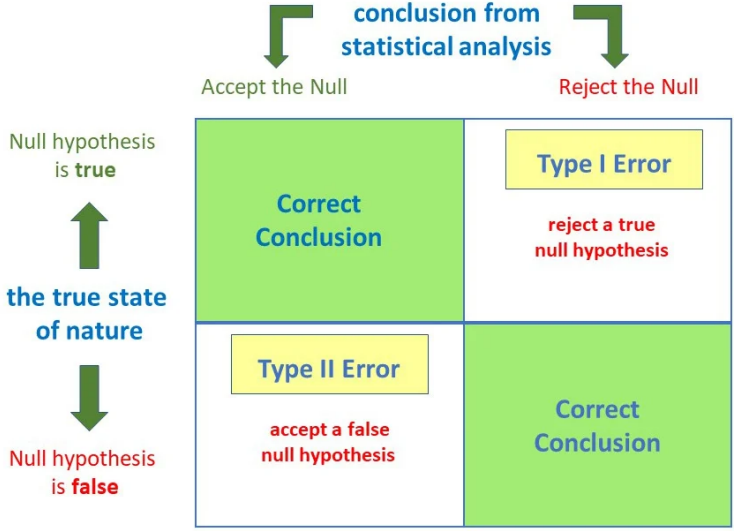
- Example: Failing to detect a real positive effect of a drug.
- Consequence: May result in missed opportunities for important discoveries or interventions.

Power

The power of a test is the probability of detecting a true effect:

$$\text{Power} = 1 - \beta.$$

Errors in Hypothesis Testing



H^0: Patient is *not* pregnant | H^1: Patient is pregnant



Heights at UPF: The Sample and Its Mean

- Suppose we collect a random sample of $n = 10$ individuals at UPF and measure their heights in centimeters. These are illustrative data.
- Let Y_i be individual i's height before sampling, and y_i its observed value.

Individual i	1	2	3	4	5	6	7	8	9	10
Height y_i (cm)	160	162	165	168	169	171	172	175	178	180

- Add the ten heights and divide by the sample size:

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i = \frac{160 + 162 + 165 + 168 + 169 + 171 + 172 + 175 + 178 + 180}{10} = 170 \text{ cm}.$$

The parameter and its estimate

The population mean height at UPF, μ_Y, is unknown. Our sample gives the estimate $\bar{y} = 170$ cm.

Heights at UPF: Standard Deviation and Standard Error

- Compute squared deviations from the sample mean, $\bar{y} = 170$:

$$\sum_{i=1}^{10}(y_i - \bar{y})^2 = (160-170)^2 + \cdots + (180-170)^2$$
$$= 100 + 64 + 25 + 4 + 1 + 1 + 4 + 25 + 64 + 100 = 388 \text{ cm}^2.$$

- Estimate the population standard deviation using $n - 1 = 9$:

$$\widehat{\sigma}_Y = s_Y = \sqrt{\frac{\sum_{i=1}^{10}(y_i - \bar{y})^2}{n-1}} = \sqrt{\frac{388}{9}} \approx 6.57 \text{ cm}.$$

- For i.i.d. observations, $\sigma_{\overline{Y}} = \sigma_Y/\sqrt{n}$. Estimate it with

$$SE(\overline{Y}) = \widehat{\sigma}_{\overline{Y}} = \frac{s_Y}{\sqrt{n}} = \frac{\sqrt{388/9}}{\sqrt{10}} \approx 2.08 \text{ cm}.$$

Interpretation

Individual dispersion: $s_Y = 6.57$ cm; sampling uncertainty: $SE(\overline{Y}) = 2.08$ cm.

Heights at UPF: Testing a Population Mean of 175 cm

- Choose $\alpha = 0.05$ and test the two-sided hypotheses:

$$H_0 : \mu_Y = 175 \text{ cm}, \text{ and } H_1 : \mu_Y \neq 175 \text{ cm}.$$

- Because $n = 10$ is small, assume heights are i.i.d. normal. Under H_0, the test statistic T follows Student's t distribution with $n - 1 = 9$ degrees of freedom.
- Using the unrounded standard error, the observed statistic is

$$t^{\text{obs}} = \frac{\bar{y} - 175}{SE(\bar{Y})} = \frac{170 - 175}{\sqrt{388/90}} \approx -2.4081.$$

- Add the probabilities in both tails of the t_9 distribution:

$$\text{p-value} = P\Big(|T| \geq |t^{\text{obs}}| \,||\, H_0\Big) = 2P(T \geq 2.4081 \mid H_0) \approx 0.0394.$$

Decision at the 5% significance level

Since $0.0394 < 0.05$, reject H_0. The sample provides evidence that the population mean height at UPF differs from 175 cm.

What are Confidence Intervals (CIs)?

Definition

A Confidence Interval (CI) is a range of values constructed from sample data that is likely to contain the true population parameter with a certain level of confidence.

Example

If we estimate the average height of a population to be 170 cm with a 95% CI to be (165 cm, 175 cm), it means we are 95% confident that the true average height falls within this interval.

- They provide a range of plausible values for a population parameter, not just a single point estimate.
- CIs help us quantify the uncertainty in our estimates.
- CIs help us make informed decisions in the presence of uncertainty.
- They are widely used in research, business, and policy to convey the reliability of estimates.

Computing CIs

- 90% confidence interval for $\mu_Y = \bar{y} \pm 1.64 \times SE(\overline{Y})$
- 95% confidence interval for $\mu_Y = \bar{y} \pm 1.96 \times SE(\overline{Y})$
- 99% confidence interval for $\mu_Y = \bar{y} \pm 2.58 \times SE(\overline{Y})$

Example

- You collect a random sample of $n = 50$ individuals at UPF and measure their heights. The observed sample mean height, $\bar{y}$, is 170 cm, and the sample standard deviation, s_Y, is 8 cm.
- The standard error, $SE(\overline{Y})$, is $\frac{8}{\sqrt{50}}$
- The 95% confidence interval is given by

$$170 \pm 1.96 \times \frac{8}{\sqrt{50}} \approx [167.78; 172.22]$$

- We are 95% confident that the true average height of adults at UPF falls within the interval [167.78 ; 172.22]

Mean Difference Testing: Comparing Groups 1/2

- Consider the null hypothesis that mean earnings for these two populations differ by a certain amount, say, d_0

$$H^0 : \mu_Y^a - \mu_Y^b = d_0 \quad \textbf{vs} \quad H^1 : \mu_Y^a - \mu_Y^b \neq d_0$$

- In a lot of applied studies $d_0 = 0$, which means

$$H^0 : \mu_Y^a = \mu_Y^b \quad \textbf{vs} \quad H^1 : \mu_Y^a \neq \mu_Y^b$$

- You have observed $\overline{y}^a$ and $\overline{y}^b$. Their sampling variance is based on the random variables $\overline{Y}^a$ and $\overline{Y}^b$.

$$\begin{aligned} var((\overline{Y}^a - \overline{Y}^b) - d_0) &= var(\overline{Y}^a) + var(\overline{Y}^b) - 2cov(\overline{Y}^a, \overline{Y}^b) \\ \Rightarrow \quad SE(\overline{Y}^a - \overline{Y}^b - d_0) &= \sqrt{\frac{var(\overline{Y}^a)}{n} + \frac{var(\overline{Y}^b)}{n}} \end{aligned}$$

Mean Difference Testing: Comparing Groups 2/2

- Once we have $SE((\mu_Y^a - \mu_Y^b) - d_0)$, we can easily compute the corresponding t-statistic

$$t = \frac{(\overline{y}^a - \overline{y}^b) - d_0}{SE((\overline{Y}^a - \overline{Y}^b) - d_0)}$$

→ We can reject that $\mu_Y^a - \mu_Y^b = d_0$ at the 5%-level when $\mid t \mid > 1.96$

- Similarly, we can construct the confidence interval for $\mu_Y^a - \mu_Y^b = d_0$

$$CI_{\mu_Y^a - \mu_Y^b = d_0} = (\overline{y}^a - \overline{y}^b) + 1.96 \times SE((\overline{Y}^a - \overline{Y}^b) - d_0)$$

→ We can reject that $\mu_Y^a - \mu_Y^b = d_0$ at the 5%-level when $d_0 \notin CI_{\mu_Y^a - \mu_Y^b = d_0}$

- Comparing mean differences is very useful, e.g., when one wants to verify that control and treatment groups are comparable in means before receiving treatment (i.e., balance tests).

Causal Inference in Randomized Control Trials

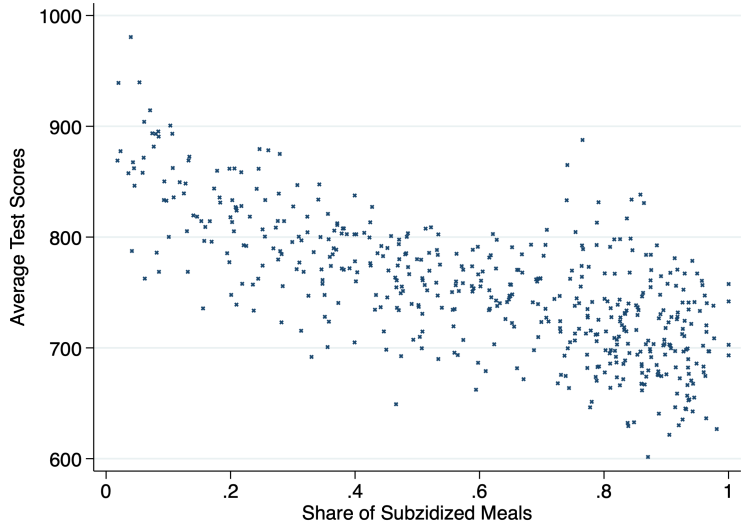
- In the ideal, golden standard RCT case, the causal effect of a treatment variable D on an outcome variable Y is the difference in conditional expectations; i.e., for a dummy treatment,

$$\mathbb{E}(Y \mid D = 1) - \mathbb{E}(Y \mid D = 0)$$

- The randomized assignment of the treatment helps ensure that any observed differences are likely due to the intervention, reducing confounding.
- Mean Difference Testing allows us to test whether $\mathbb{E}(Y \mid D = 1) = \mathbb{E}(Y \mid D = 0)$
- Example 1: Drug Efficacy
  - Treatment Group: Receives the new drug.
  - Control Group: Receives a placebo.
  - Mean Difference Testing helps determine if the drug has a significant effect on health outcomes.
- Example 2: Educational Interventions
  - Different teaching methods in classrooms.
  - Mean Difference Testing assesses if one method leads to better student performance.

The California Test Score Data Set

- All K-6 and K-8 California school districts ($n = 500$) (CPS, 1999)
- Test score: sum of math and English/language arts, 5th grade
- Subsidized Meals: free or reduced-price meals



The California Test Score Data Set

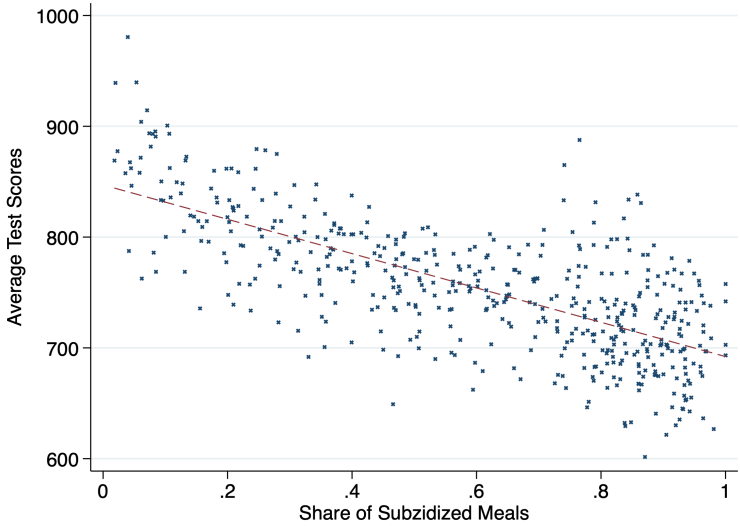
The scatter plot does not tell us much about each variable, independently. We can provide descriptive statistics of the sample:

	Median	p25	p75	Min	Max
Test Scores	751.5	708.95	791.75	601.7	980.7
Share of Subsidized Meals	.6511	.3755	.8432	.018	1

	Mean	sd	Skewness	Kurtosis
Test Scores	753.8542	60.3035	.3696458	3.150585
Share of Subsidized Meals	.6018114	.2758199	-.445432	1.958609

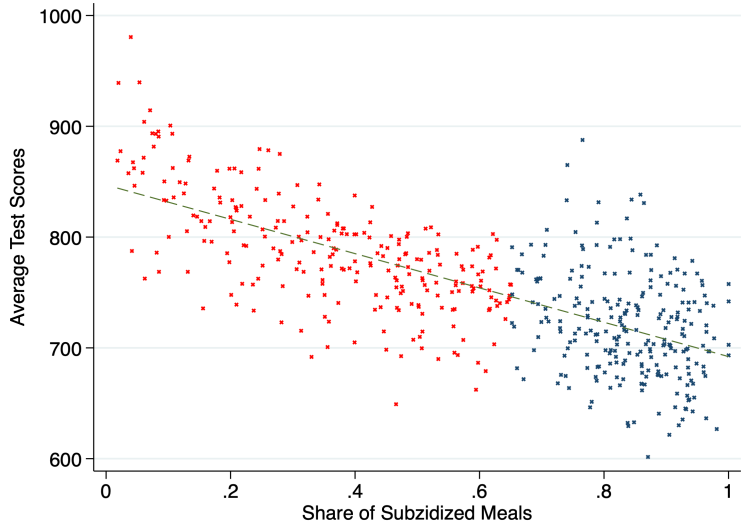
The California Test Score Data Set

- This allows us to study the joint distribution of the *Test Scores* and the *Share of Subsidized Meals*
- We can compute the sample covariance, $s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})$
- Here, $s_{XY} = -11.7839$
- The sample correlation, $r_{XY} = \frac{s_{XY}}{s_X s_Y}$
- Here, $r_{XY} \approx \frac{-11.78}{60.3 \times 0.28} \approx -.7$



The California Test Score Data Set

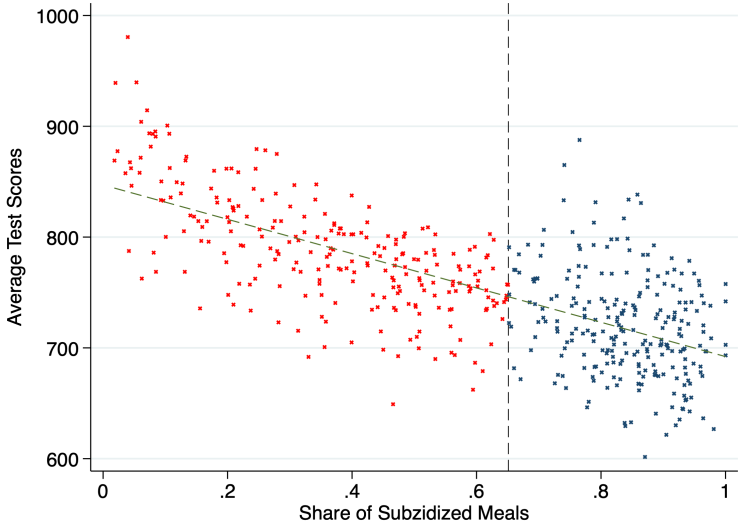
- Are schools with less subsidized meals significantly better than schools with more subsidized meals?
- We can split the distribution into two groups:
  1. Below-median *Share of Subsidized Meals* (red)
  2. Above-median *Share of Subsidized Meals* (red)



The California Test Score Data Set

Estimation:

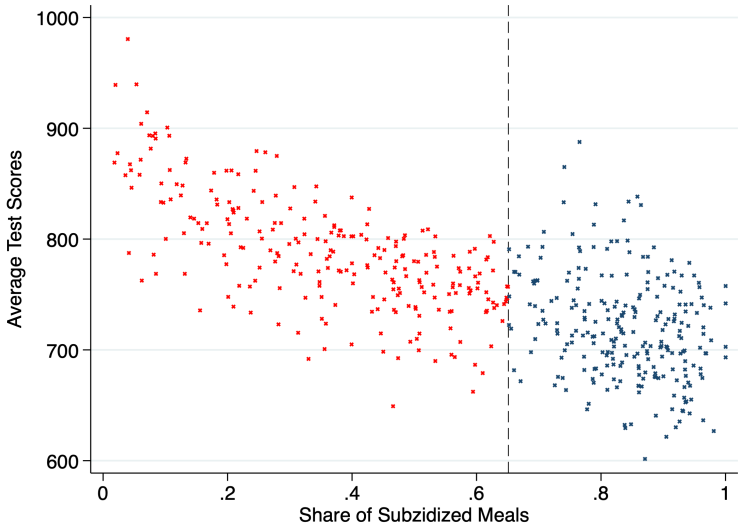
- Compare average test scores in districts with low *Share of Subsidized Meals* to those with high *Share of Subsidized Meals*



The California Test Score Data Set

Hypothesis testing:

- Test the "null" hypothesis that the mean test scores in the two types of groups are the same, against the "alternative" hypothesis that they differ
- $H_0 : \mu_Y^{X^L} - \mu_Y^{X^H} = 0$
- $H_1 : \mu_Y^{X^L} - \mu_Y^{X^H} \neq 0$

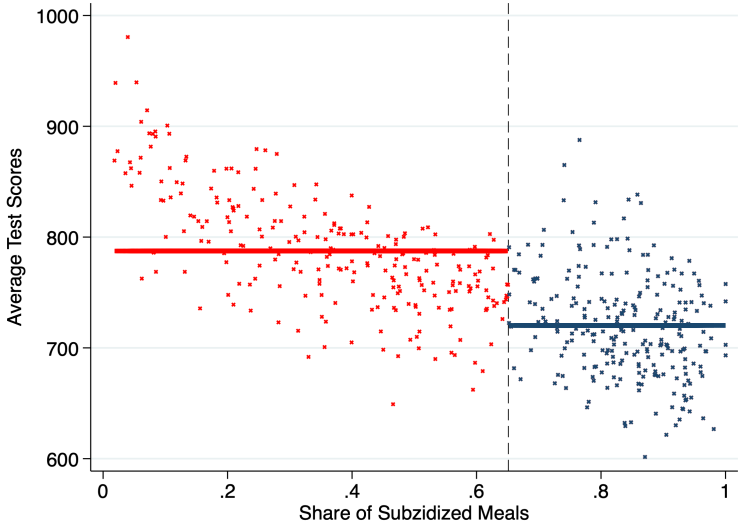


The California Test Score Data Set

We compute the descriptives for each group

	$\bar{y}$	s^2	n
X^L	787.46	52.62	250
X^H	720.24	47.44	250

- $\bar{y}^{X^L} - \bar{y}^{X^H} = 67.214$
- $SE(\overline{Y}^{X^H} - \overline{Y}^{X^L}) = \sqrt{\frac{47.44^2}{250} + \frac{52.62^2}{250}} \approx 4.48$
- $t = 67.214/4.48 \approx 15 > 1.96$

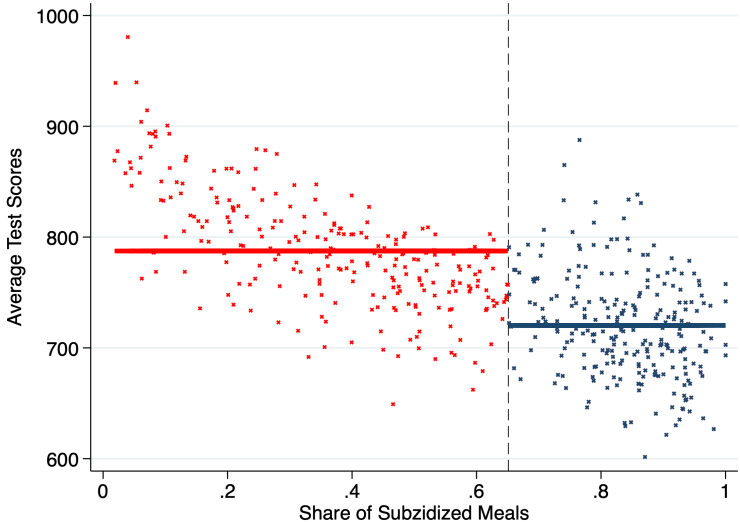


The California Test Score Data Set

Alternatively, we can also compute the 95% CI for the difference

$$\begin{aligned}\bar{y}^{X^L} - \bar{y}^{X^H} &\pm 1.96 \times SE(\overline{Y}^{X^H} - \overline{Y}^{X^L}) \\ &= 67.214 \pm 1.96 \times 4.48 \\ &= [58.41; 76.018] \\ &\notin 0\end{aligned}$$

-



Material I

- **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapter 3